

VARUNKUMAR SONAWANE

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Summary

AI-focused Software Engineer with a growth mindset and hands-on experience building **data pipelines** and scalable backend systems, with strong data structures, problem solving, analytical reasoning, and cross-functional collaboration, integrating **AI agents and LLM-based automation** on **AWS** and **GCP** using **Python** and **SQL**.

Education

Indiana University Bloomington, IN, USA.

Master of Science in Computer Science

Aug 2024 – May 2026

CGPA: 3.83/4

Certifications

AWS Certified Developer – Associate | Microsoft Azure AI Fundamentals

Experience

Graduate Teaching Assistant, Indiana University, Bloomington, IN

Dec 2025 - Present

- Mentored **120+** students on core software engineering topics, providing detailed feedback on **code quality, documentation, and object-oriented design**.
- Guided teams through the full **software development life cycle (SDLC)** including requirements, design, implementation, testing, deployment, and debugging of distributed systems, NLP pipelines, and web scraping applications in Python, Scala, and Spark.
- Collaborated** with the instructor to develop assignments, project rubrics and coding labs on **SQL, version control, Agile workflows, and Tableau dashboards**, strengthening end-to-end engineering practices.

Data Engineer Intern, The Commons XR, San Diego, CA

May 2025 - Aug 2025

- Designed and implemented scalable backend pipelines to migrate and validate **5M+ records** per week from PostgreSQL to BigQuery, eliminating manual handoffs and improving system reliability.
- Created and maintained **10+ production** features and services in **Python, Plotly, and Dash** to support real-time and **ad-hoc** analytics for XR platforms.
- Integrated an **LLM-powered automation** service on Vertex AI to replace manual post-session analysis, generating structured insights and reducing recurring engineering effort.

Software Engineering Intern, CodeClause, India

July 2023 - Jan 2024

- Increased backend system reliability by reducing recurring service failures across **5+ microservices** by implementing unit tests, improving error handling, and integrating changes through a **CI/CD pipeline**.
- Delivered a **high-impact** user-facing feature by building and integrating a **React-based frontend** with REST APIs, improving feature usability and reducing manual support requests.
- Reduced resolution time by **~20%** by defining infrastructure using **Infrastructure as Code (IaC)** and adding structured **logging and metrics**, enabling faster root-cause analysis.

Projects

VoiceLegal-AI Web Extension

- Built a **conversational AI** system for legal document analysis by integrating a web client, backend services, and Chrome extension, delivering **4.8x faster** responses than traditional AI tools for real-time question answering.
- Elevated user experience** by integrating **Vertex AI (Gemini 2.5)** and **ElevenLabs voice**, delivering explainable natural-language responses through chat and voice.

Cloud-Based Web Scraping & Analytics Pipeline | *Glue, Athena SQL, DynamoDB, Power BI, SNS, Docker*

- Automated daily ingestion of 10K+ records using containerized Scrapy jobs on ECS Fargate and Amazon S3.
- Build a cloud analytics pipeline with AWS Glue, Athena, **DynamoDB, and Lambda** for daily dashboards and alerts.

IdeaGenie - AI-Driven Idea Evaluation Platform | *Luddy Hackathon Winner*

- Developed a **full-stack AI** product by developing a Flask REST API and a React frontend to evaluate and rank top 3 user ideas, enabling automated prioritization and reducing manual review effort.
- Improved decision quality and usability** by integrating LLM-based reasoning with an interactive UI, allowing users to adjust parameters and receive explainable, real-time feedback.

Technical Skills

Languages: Python, SQL, Bash, HTML, CSS | **Systems:** REST APIs, Microservices, CI/CD, Linux | **Databases:** PostgreSQL, MySQL, MongoDB, GraphQL, Cassandra, Neo4j | **Cloud:** AWS, GCP, Azure, Docker, Jenkins, Kubernetes | **AI/ML:** TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, NLTK | **Distributed:** Spark, Airflow | **Tools:** Git, Excel